

Crosstalk-Aware Feedforward Drive for Fast Reconfiguration of Thermo-Optic Programmable Photonic Meshes

Md. Al Mohaimin Bhuiyan Arfin

Abstract—Programmable photonic meshes in silicon are tuned by thermo-optic heaters whose heat also reaches neighbouring waveguides. Existing compensation corrects this thermal crosstalk only in steady state. We show that its transient, not the heater’s own response, limits how fast a mesh can be reconfigured to high accuracy, and we remove most of it with drive waveforms designed for all heaters jointly. A three-dimensional finite-volume thermal model of an 8×8 silicon-on-insulator Clements mesh with 56 heaters is solved exactly in time by separation into 2.6×10^5 decoupled modes, validated against a direct solver, checked for grid convergence and compared with published heater measurements. It shows crosstalk of only 0.8–1.8% per neighbour but with 63% rise times of 79–286 μs and substrate time constants up to 2.4 ms. A compact shared-pole model identified from heater step responses reproduces it to within 0.16 mrad of optical phase. Using this model, the voltage waveforms of all heaters are optimised by quasi-Newton descent on the exact, nonlinear matrix error, with the heaters’ temperature-dependent resistance and the driver’s voltage limit included; every design ends at the usual static calibration. Evaluated on the three-dimensional model over 16 random reconfigurations, the method shortened the median settling time from 131 to 59 μs to a 10^{-2} matrix error and from 940 to 201 μs to 10^{-3} , in every case, whereas per-heater pre-emphasis changed them by less than 4%. We also quantify the effect of model order, driver headroom, fabrication variations and converter resolution. The method needs no change to the photonic chip.

Index Terms—Programmable photonics, thermo-optic phase shifter, thermal crosstalk, Mach–Zehnder interferometer mesh, feedforward control, model order reduction, silicon photonics, optical neural networks.

I. INTRODUCTION

PROGRAMMABLE photonic integrated circuits built from meshes of Mach–Zehnder interferometers (MZIs) implement arbitrary linear transformations of optical fields [1]–[3]. They are the core of general-purpose photonic signal processors [4], [5], optical neural networks and matrix accelerators [6]–[8], and reconfigurable switches and filters. In silicon photonics the phase of each MZI is almost always set by a thermo-optic phase shifter: a resistive heater above the waveguide that changes the refractive index through the thermo-optic effect [9]–[11]. Heaters are simple, lossless and compatible with any foundry, but they are slow (microseconds) and they are not independent. Heat spreads through the cladding, the buried oxide and above all the silicon substrate, so every heater also

shifts the phase of its neighbours. This thermal crosstalk grows with integration density and with the number of heaters that are on at once.

Thermal crosstalk has so far been treated as a *static* problem. Measured or simulated steady-state coupling matrices are inverted so that, once the chip has settled, every programmable phase takes its target value [12]–[15]. Design measures such as trenches, suspended waveguides and optimised heater placement reduce the coupling itself [10], [11], and commercial circuit simulators now model it through a steady-state coupling matrix [16]. Static compensation fixes the final state, but it says nothing about the path to it. After a reconfiguration command the heated waveguide responds within microseconds, while the heat reaching distant waveguides through the substrate arrives tens to thousands of microseconds later. Until that slow heat has settled the mesh implements the wrong matrix. For applications that reprogram the mesh often, such as updating the weights of an optical neural network, time-multiplexed matrix operations or circuit switching, this settling time rather than the heater’s own time constant sets the usable reconfiguration rate.

The transient of a single heater can be shortened by pre-emphasis: overdriving the heater briefly before settling at its final power [17]. Applied heater by heater, however, pre-emphasis ignores crosstalk. As shown below, it barely changes the settling time of a mesh, because what limits settling is the slow heat exchanged *between* heaters. Lumped transient thermal models of heater arrays have been reported [13], but they were used to design driving schemes for a single array rather than to compute crosstalk-aware drive waveforms for a whole mesh. Shaping command inputs to suppress unwanted dynamics is a classical idea in motion control [18], and gradient-based optimisation is established for programming mesh phases in steady state [19] and, through differentiable simulators, for co-designing photonic and electronic circuits [20]. To our knowledge it has not been used to shape heater drives against transient thermal crosstalk.

This paper treats reconfiguration as a multi-input dynamic problem and makes four contributions.

- 1) A three-dimensional finite-volume thermal model of an 8×8 thermo-optic Clements mesh on silicon-on-insulator, solved *exactly in time* by separation into 2.6×10^5 decoupled modes. It is validated against a direct solver, checked for grid convergence and compared with published heater measurements. It quantifies how slow and how far-reaching thermal crosstalk is.

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TABLE I
MODEL PARAMETERS

Quantity	Value
Si substrate: $k, \rho c$	148 W/(m K), 1.64 MJ/(m ³ K)
SiO ₂ (BOX, cladding): $k, \rho c$	1.38 W/(m K), 1.64 MJ/(m ³ K)
Si thermo-optic coefficient	$1.86 \times 10^{-4} \text{ K}^{-1}$
Wavelength	1.55 μm
TiN heater: R_0, α_R	500 Ω , $1 \times 10^{-3} \text{ K}^{-1}$
Driver output resistance R_s	50 Ω
Driver voltage limit V_{max}	8 V (6.5 V to 10 V in Sec. VI-D)

- 2) A compact shared-pole model, identified from heater step responses, that reproduces the three-dimensional model to within 0.16 mrad of optical phase while being cheap and differentiable.
- 3) A feedforward design method: drive waveforms for all heaters are optimised jointly, by gradient-based quasi-Newton optimisation, against the exact (nonlinear) matrix error of the mesh. The method includes the heater's temperature-dependent resistance and the driver's voltage limit, and every design ends exactly at the static calibration.
- 4) An evaluation in which every result is computed on the three-dimensional model, not on the model used for design. It covers many random reconfigurations, comparison with static compensation and per-heater pre-emphasis, driver headroom, model order and fabrication variations.

The method is simulation-based, but it uses only quantities a lab can obtain (heater step responses and the static calibration that programmable meshes already require), and it runs from pre-computed waveforms, so it needs no change to the photonic chip.

Section II describes the thermal model and its validation. Section III formulates reconfiguration and the error metric, Section IV the compact model, and Section V the drive design. Results are in Section VI, and limitations are discussed in Section VII.

II. THERMAL MODEL OF A THERMO-OPTIC MESH

A. Device, Layout and Stack

We consider an $N \times N$ Clements mesh [2] with $N = 8$: N columns, $M = N(N - 1)/2 = 28$ MZIs and one external and one internal heater per MZI, $H = 2M = 56$ heaters in total (Fig. 1). Waveguides are 30 μm apart and each column is 300 μm long. Heaters are 100 μm long, lie on the upper arm of each MZI, and sit centred at 15% (external) and 55% (internal) of the column length; the two 50:50 couplers of each column sit at 35% and 75%. The stack is a standard silicon-on-insulator platform: a 725 μm silicon substrate held at constant temperature at its back side, a 2 μm buried oxide and a 3 μm oxide cladding, with TiN heaters 1.75 μm above the waveguide plane. Material properties are listed in Table I. The 220 nm silicon device layer is thermally negligible at this scale and is treated as oxide.

B. Finite-Volume Model and Exact Modal Solution

The temperature rise $T(\mathbf{r}, t)$ obeys the heat equation with the heater powers as sources. It is discretised by cell-centred finite volumes on a tensor-product grid with harmonic-mean face conductances. The grid has 25 μm cells along the propagation direction x ; 2.5 μm cells within 7.5 μm of every waveguide and 5 μm cells between waveguides in y ; and 0.5 μm cells in the oxides, growing geometrically (ratio 1.6) through the substrate in z . Lateral boundaries 300 μm from the mesh and the top surface are adiabatic. This gives $120 \times 87 \times 25 = 261\,000$ cells and the semi-discrete system

$$\mathbf{C} \dot{\mathbf{T}}(t) = -\mathbf{G} \mathbf{T}(t) + \mathbf{Q} \mathbf{P}(t), \quad (1)$$

with diagonal heat capacities $\mathbf{C}$, symmetric positive-definite conductance matrix $\mathbf{G}$ and heater-to-cell distribution matrix $\mathbf{Q}$ acting on the heater powers $\mathbf{P} \in \mathbb{R}^H$.

Because the material properties vary only with z and the grid is a tensor product, (1) separates exactly. The x -direction operator is the uniform Neumann Laplacian, diagonalised by discrete cosine modes ϕ_p^x with eigenvalues μ_p . The weighted y -direction operator $\mathbf{L}_y$ is diagonalised by the generalised eigenvectors $\mathbf{L}_y \phi_q^y = \nu_q \mathbf{D}_y \phi_q^y$, with $\mathbf{D}_y$ the diagonal of cell widths. For every pair (p, q) what remains is a one-dimensional problem in z ,

$$(\mathbf{L}_z + (\mu_p + \nu_q) \mathbf{K}_z) \psi_{pqr} = \sigma_{pqr} \mathbf{C}_z \psi_{pqr}, \quad (2)$$

where $\mathbf{L}_z$ holds the vertical conductances and the isothermal bottom, and $\mathbf{K}_z$ and $\mathbf{C}_z$ hold the layer conductivities and heat capacities multiplied by the cell thicknesses. Each of the 261 000 modal amplitudes c_{pqr} then obeys

$$\dot{c}_{pqr} = -\sigma_{pqr} c_{pqr} + \mathbf{b}_{pqr}^\top \mathbf{P}(t), \quad (3)$$

which is integrated *exactly* for piecewise-constant powers over any time step:

$$c_{pqr}(t_{n+1}) = e^{-\sigma_{pqr} h_n} c_{pqr}(t_n) + \frac{1 - e^{-\sigma_{pqr} h_n}}{\sigma_{pqr}} \mathbf{b}_{pqr}^\top \mathbf{P}_n. \quad (4)$$

The model has no time-discretisation error and needs no linear solves. A 10 ms transient on 1191 time steps, including the heater nonlinearity, takes about 4.5 s on one CPU core, against about three minutes for a 40 μs transient with the direct solver.

The optical phase of each waveguide interval j (between consecutive couplers of the same waveguide) is

$$\varphi_j(t) = \frac{2\pi}{\lambda} \frac{dn}{dT} \int_{x \in j} T(x, y_j, 0, t) dx, \quad (5)$$

evaluated for all $2N^2 = 128$ intervals, heated or not, so crosstalk enters the mesh transfer matrix exactly. Each heater's own temperature is also computed, since its resistance depends on it (Section III).

C. Validation

The modal solution was checked against a direct solution of (1), using algebraic-multigrid-preconditioned conjugate gradients [21]. The steady states agree to a relative 2.7×10^{-14} . Over a 40 μs transient, a variable-step second-order backward-differentiation solver with 0.25 μs steps differs from the exact

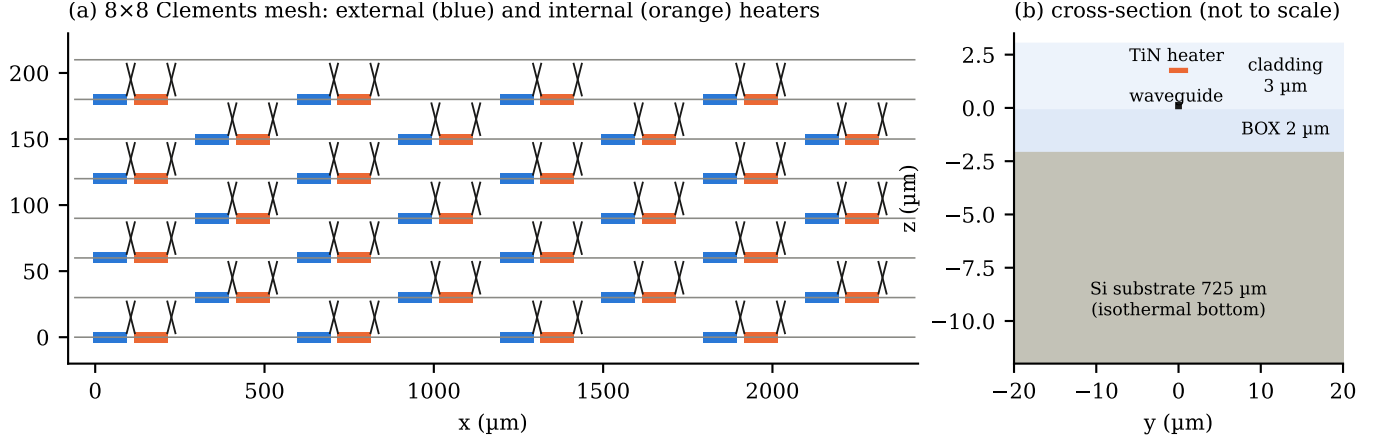


Fig. 1. (a) Layout of the 8×8 Clements mesh: waveguides (grey), 50:50 couplers (crosses), external (blue) and internal (orange) heaters. (b) Cross-section of the silicon-on-insulator stack.

modal solution by at most 0.5% of the final value, which is that solver's own time-discretisation error. Refining the y and z grids by a factor of two, or the x grid by a factor of two, changes P_π by at most 2.2%, the crosstalk ratios by at most 2.1% and the self-heating time constant by at most 3%.

The model gives $P_\pi = 25.8 \text{ mW}$ for the MZI differential phase and a self-heating 63% rise time of $8.5 \mu\text{s}$. Jacques *et al.* measured $P_\pi = 21.4 \text{ mW}$ and $\tau = 5.6 \mu\text{s}$ for TiN heaters on a comparable silicon-on-insulator platform [10], so the model is realistic without any fitted parameter.

D. Transient Thermal Crosstalk

Fig. 2 shows the phase response of the heated waveguide and of waveguides 30, 60 and $90 \mu\text{m}$ away to a heater step. The steady crosstalk is small, 1.83%, 1.12% and 0.79% of the heated waveguide's phase, and it decays slowly with distance, because most of it is carried by the substrate. It is also slow: 63% rise times of 79, 156 and $286 \mu\text{s}$ against $8.5 \mu\text{s}$ for the heated waveguide, with a slowest thermal mode of 2.4 ms. Two properties matter for reconfiguration. First, with 56 heaters dissipating about 1.1 W in total, these small coefficients add up: ignoring crosstalk leaves a steady matrix error of 0.08 in the example of Fig. 3. Second, the slow part is also present in each heater's own response, which reaches 99% of its final value after $65 \mu\text{s}$ but 99.9% only after 1.56 ms. Settling to high accuracy is therefore governed by substrate heat that is shared among all heaters.

III. RECONFIGURATION PROBLEM

A. Electrical Drive

Each heater h is driven by a voltage $V_h(t)$ through the driver's output resistance R_s . Its resistance rises with its own temperature T_h , so the dissipated power is

$$P_h(t) = \frac{V_h(t)^2 R_h(t)}{(R_h(t) + R_s)^2}, \quad R_h(t) = R_0(1 + \alpha_R T_h(t)), \quad (6)$$

with $0 \leq V_h \leq V_{\text{max}}$. The heater's own temperature rises by about 4 K per milliwatt in the thermal model, so at 50 mW the

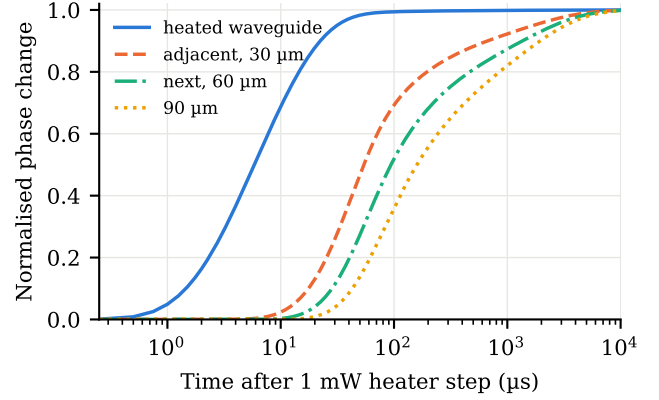


Fig. 2. Normalised phase response of the heated waveguide and of waveguides 30, 60 and $90 \mu\text{m}$ away to a heater step (three-dimensional model). Steady crosstalk ratios: 1.83%, 1.12% and 0.79%.

resistance rises by about 20% and (6) is appreciably nonlinear. All simulations in this paper include (6).

B. Mesh Transfer Matrix and Static Calibration

The $N \times N$ transfer matrix $\mathbf{U}(\varphi)$ follows from the interval phases (5) and ideal 50:50 couplers, column by column. The programmable phase of heater h is the difference θ_h between the phases of the upper and lower arms at its location (external or internal slot of its MZI). A target configuration is a vector θ^* of H such phases. Static calibration, the state of the art [12]–[14], uses the steady-state gains $\mathbf{K} \in \mathbb{R}^{H \times H}$ from heater powers to programmable phases and sets

$$\mathbf{P}_{\text{ss}} = \mathbf{K}^{-1} \theta^*, \quad (7)$$

adding 2π to any target whose power would be negative. The voltages $\mathbf{V}_{\text{ss}}$ follow from (6) with the steady heater temperatures. Using only the diagonal of $\mathbf{K}$ ignores crosstalk; we call this the *naïve* calibration. The target matrix $\mathbf{U}_{\text{ss}}$ is the one reached in steady state with $\mathbf{P}_{\text{ss}}$. It includes the heat deposited in unheated waveguide intervals, so it is exactly the matrix a calibrated chip settles to.

C. Reconfiguration and Error Metric

A reconfiguration starts from the steady state of one calibrated configuration A and applies drive waveforms $\mathbf{V}(t)$ that end at $\mathbf{V}_{ss}$ of configuration B. Its accuracy at time t is measured by

$$e(t) = \frac{1}{\sqrt{N}} \min_{\mathbf{D}} \|\mathbf{D} \mathbf{U}(t) - \mathbf{U}_{ss}\|_F, \quad (8)$$

where $\mathbf{D}$ ranges over diagonal unitary matrices. The minimisation removes the phases at the mesh outputs. The $H = N(N - 1)$ heaters of a Clements mesh cannot set them, and they do not affect intensity detection, but every other deviation of the matrix counts. The minimum has the closed form $D_{ii} = \langle \mathbf{u}_i^{ss}, \mathbf{u}_i \rangle / |\langle \mathbf{u}_i^{ss}, \mathbf{u}_i \rangle|$ for rows $\mathbf{u}_i$. The settling time t_ε is the time after which $e(t) \leq \varepsilon$ for the rest of a 10 ms horizon; we report $\varepsilon = 10^{-2}$ and 10^{-3} , together with the integrated error $\int_0^{10 \text{ ms}} e(t) dt$. Reconfigurations use random targets: external phases uniform in $[0, 2\pi)$ and internal phases uniform in $[0, \pi]$.

IV. COMPACT THERMAL MODEL

Drive design needs a thermal model that is cheap and differentiable. The three-dimensional model is used only to produce the heater step responses a designer would measure or simulate, and afterwards as the “chip” on which designs are evaluated. For design we use a shared-pole model: for every output o (the 128 interval phase integrals and the 56 heater temperatures) and heater h ,

$$y_o(t) = \sum_{h=1}^H \sum_{k=1}^K R_{ohk} x_{hk}(t), \quad \tau_k \dot{x}_{hk} = -x_{hk} + P_h(t). \quad (9)$$

This is a common-pole rational model, as used in vector fitting [22], and a simple form of model order reduction [23]. The time constants τ_k are fixed on a logarithmic grid from $0.25 \mu\text{s}$ to the slowest thermal time constant, 2.4 ms . The residues R_{ohk} follow from one linear least-squares fit to the sampled step responses, with samples weighted uniformly in logarithmic time and the steady-state gains enforced exactly ($\sum_k R_{ohk}$ equals the static gain). With piecewise-constant powers, (9) is integrated exactly in the same way as (4). It has HK states against 2.6×10^5 modes.

Table II shows the accuracy against model order. With $K = 24$ the largest step-response error is 0.003% of the heated waveguide’s response, and during a full reconfiguration the model’s interval phases differ from the three-dimensional model’s by at most 0.16 mrad; its predicted matrix error is indistinguishable from the reference down to 10^{-5} . Lower orders leave errors of several milliradians. Section VI-C shows what this costs.

V. DRIVE DESIGN

A. Parametrisation

The drive of heater h is written as

$$V_h(t) = V_{\max} \sigma \left(\logit \frac{V_h^b(t)}{V_{\max}} + \sum_{b=1}^B z_{bh} g_b(t) \right), \quad (10)$$

TABLE II
ACCURACY OF THE COMPACT MODEL AGAINST THE
THREE-DIMENSIONAL MODEL

K	States	Max. step error (% of self)	Max. phase error in reconfiguration (mrad)
8	448	0.405	24.42
12	672	0.059	3.59
16	896	0.035	2.07
24	1344	0.003	0.16

Phase error: largest interval-phase difference over three static-step reconfigurations, 10 ms.

where σ is the logistic function and V_h^b is a starting waveform (below). The form keeps $0 < V_h < V_{\max}$ for any parameters z_{bh} , so the driver limit never has to be imposed as a constraint. The basis functions g_b are 64 piecewise-linear hat functions, spaced $1 \mu\text{s}$ up to $40 \mu\text{s}$, $10 \mu\text{s}$ up to $200 \mu\text{s}$ and $100 \mu\text{s}$ up to 1 ms , and one decaying exponential e^{-t/τ_k} for each identified time constant $\tau_k \geq 2 \mu\text{s}$. Every basis function vanishes as t grows, so every design converges to the static calibration $\mathbf{V}_{ss}$. The exponentials let a heater cancel, with the right time course, slow heat that arrives from its neighbours. With $K = 24$ there are $B = 83$ functions and $BH = 4648$ design variables.

B. Objective

Settling spans five decades of time and error, from $e \approx 1$ in the first microseconds to 10^{-3} after hundreds of microseconds. A plain squared error would ignore the tail. We therefore weight every decade of time equally and measure the error relative to static compensation:

$$J(\mathbf{z}) = \sum_n w_n e(t_n; \mathbf{z})^2, \quad w_n \propto \frac{\Delta \ln t_n}{\bar{e}(t_n)^2 + \delta^2}, \quad (11)$$

where $\bar{e}$ is the error of static compensation predicted by the compact model and $\delta = 10^{-4}$ keeps the weights finite once $\bar{e}$ is negligible. J is evaluated on the exact nonlinear error (8), through the compact model (9), the heater nonlinearity (6) and the mesh transfer matrix, over the full 10 ms horizon (1191 time steps).

C. Optimisation

Every step above is differentiable, so the gradient $\nabla_{\mathbf{z}} J$ follows by reverse-mode automatic differentiation [24]. We minimise J with L-BFGS [25] (history 30, strong-Wolfe line search), five restarts of 20 iterations each. One evaluation of J and its gradient takes about 0.8 s on one CPU core, and a design about 100 s. Starting from $\mathbf{z} = \mathbf{0}$ means starting from the starting waveform V^b . We found that starting from per-heater pre-emphasis (below), rather than from the static step, gives a lower objective and a faster early transient. The proposed design is thus a coordinated, multi-heater correction added to per-heater pre-emphasis.

A convex quadratic-programming variant, which linearises (8) around the target, was also tried. It was abandoned: the linearised error is blind to phase deviations that do not change $\mathbf{U}$ to first order, and the optimum exploited them with large excursions that cost accuracy at second order. Optimising the exact error avoids this.

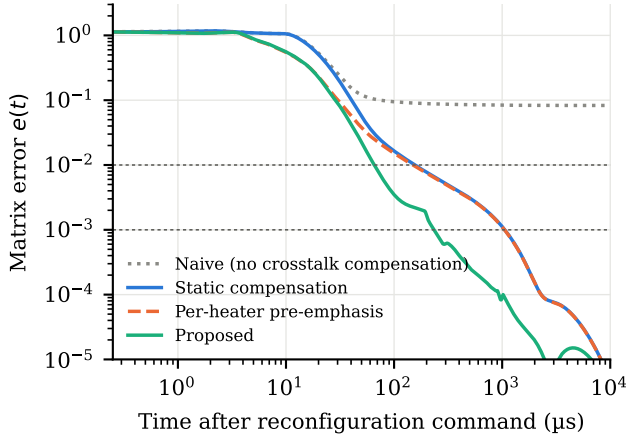


Fig. 3. Matrix error $e(t)$ after one random reconfiguration of the 8×8 mesh, computed on the three-dimensional model. Dotted lines: 10^{-2} and 10^{-3} .

D. Baselines

Three baselines are evaluated on the same reconfigurations. *Naive*: a voltage step to the naive calibration. *Static*: a voltage step to V_{ss} , i.e. static crosstalk compensation. *Per-heater pre-emphasis*: each heater is driven at V_{max} (if its power rises) or at zero (if it falls) for a time t_h^{on} , then held at $V_{ss,h}$. Each t_h^{on} is chosen to bring that heater's own programmable phase to its target as fast as possible, using its own identified response. It follows the fast-switching drive schemes demonstrated for single thermo-optic switches [17], applied to every heater independently; crosstalk is compensated only statically.

VI. RESULTS

All results in this section are computed on the three-dimensional model of Section II, including the heater nonlinearity (6). Designs use only the compact model of Section IV. Unless stated otherwise $K = 24$ and $V_{max} = 8$ V. The static calibrations need at most 51.6 mW and 6.04 V per heater and about 1.07 W in total.

A. An Example Reconfiguration

Fig. 3 shows the matrix error after one random reconfiguration. Without crosstalk compensation the mesh never settles: the error stays at 0.08. Static compensation removes this floor, but the error falls below 10^{-2} only after 156 μ s and below 10^{-3} only after 1.06 ms, following the slow substrate heat. Per-heater pre-emphasis speeds up the first tens of microseconds (error 0.23 against 0.51 at 20 μ s) but leaves the tail, and therefore both settling times, essentially unchanged (150 μ s and 1.06 ms). The proposed drive keeps the pre-emphasis gain and also cuts the tail: it reaches 10^{-2} after 65.8 μ s and 10^{-3} after 235 μ s, 4.5 times sooner than static compensation.

Fig. 4 shows where the gain comes from. The pre-emphasis pulses (Fig. 4a) are almost identical in both methods and last a few microseconds (about 5 μ s per heater on average). What the proposed design adds is a small, slowly decaying correction on the heaters after their pulses (Fig. 4b): at most 11.6 mV in this

TABLE III
RESULTS OVER 16 RANDOM RECONFIGURATIONS (THREE-DIMENSIONAL MODEL): MEDIAN [RANGE]

Strategy	$t_{10^{-2}}$ (μ s)	$t_{10^{-3}}$ (μ s)	$\int e dt$ (μ s)
Naive	never	never	835 [662, 1039]
Static compensation	131 [71, 360]	940 [335, 1680]	27.5 [21.1, 39.0]
Per-heater pre-emphasis	126 [64, 355]	935 [330, 1680]	18.6 [13.0, 28.0]
Proposed	59 [50, 93]	201 [101, 550]	13.9 [10.6, 17.9]

Naive: steady error 0.082 [0.065, 0.101]. t_e : time after which $e(t) \leq \epsilon$ within 10 ms.

example (4.8 to 34 mV across all reconfigurations), decaying over about a millisecond. Each heater thereby cancels, as it arrives, the heat that its neighbours' changes send through the substrate. The correction is a fraction of a percent of the drive voltage, but because the slow heat is common to all heaters it can only be found by designing all heaters together.

B. Statistics Over Random Reconfigurations

Table III and Fig. 5 summarise 16 random reconfigurations. The proposed drive reduced the median settling time to 10^{-2} from 131 μ s to 59.2 μ s (a factor of 2.24, range 1.31 to 3.93) and to 10^{-3} from 940 μ s to 201 μ s (a factor of 4.33, range 2.39 to 7.74). It was faster than static compensation at both thresholds in all 16 cases. It also made the settling time far more predictable: the range of settling times to 10^{-3} shrank from 335–1680 μ s to 101–550 μ s. Per-heater pre-emphasis lowered the integrated error by a third but changed the median settling times by less than 4%, confirming that settling is limited by crosstalk rather than by each heater's own response.

The settling times predicted by the compact model agreed with those computed on the three-dimensional model to within 1 μ s in every case. A design took about 100 s (121 evaluations of J and its gradient) on one CPU core.

C. Compact-Model Order

The design is only as good as the model it is designed on. Table IV repeats six reconfigurations with lower-order compact models, again evaluated on the three-dimensional model. With $K = 12$ (largest phase error 3.6 mrad) the drive still halves the settling time to 10^{-2} , but its median advantage at 10^{-3} disappears (faster in 4 of 6 cases, median unchanged). With $K = 8$ (24 mrad) it is no faster than static compensation at 10^{-2} and much slower at 10^{-3} (5.6 ms against 0.74 ms). The model itself does not reveal this: with $K = 8$ it predicted 239 μ s to 10^{-3} . The phase error of the design model must therefore lie well below the phase error that corresponds to the target accuracy, roughly a few milliradians for 10^{-2} and a fraction of a milliradian for 10^{-3} . With the logarithmic pole grid used here, $K = 24$ poles (1344 states) meet this requirement.

D. Driver Headroom

The largest static voltage in these reconfigurations is 6.04 V. Lowering the driver limit from 8 to 6.5 V, which leaves almost no room for overdrive on the hottest heaters, or raising it to

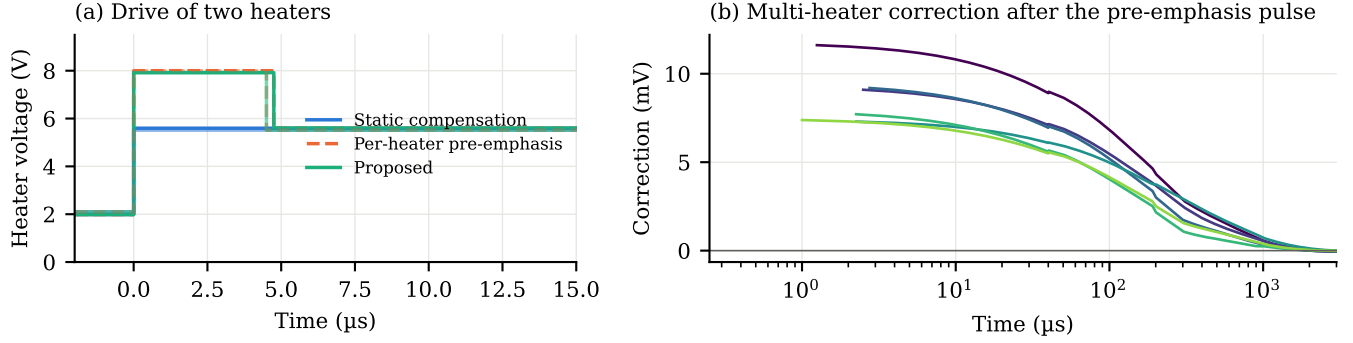


Fig. 4. Drive waveforms for the reconfiguration of Fig. 3. (a) The two heaters with the largest voltage change. (b) The difference between the proposed drive and per-heater pre-emphasis after each heater's pulse, for the six heaters with the largest correction.

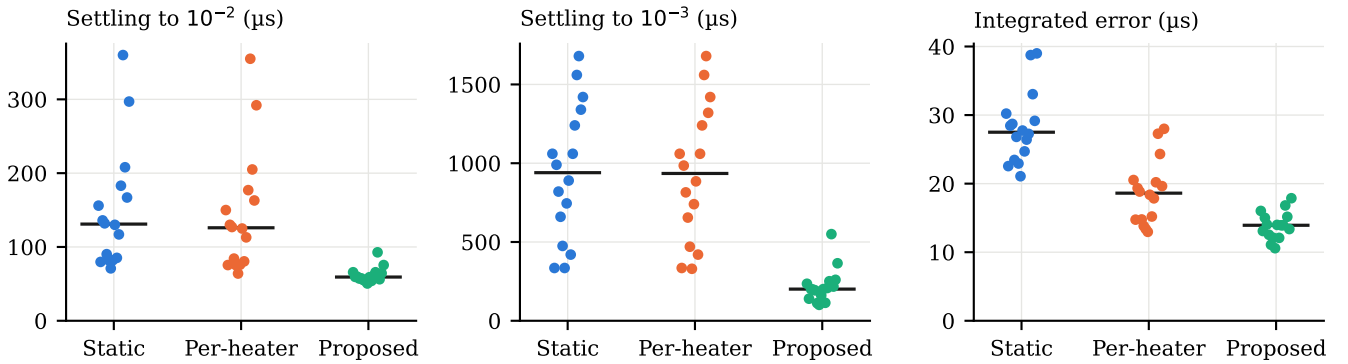


Fig. 5. Settling times and integrated error for 16 random reconfigurations (dots), with medians (bars), evaluated on the three-dimensional model.

TABLE IV
SENSITIVITY STUDIES: MEDIAN OF 6 RECONFIGURATIONS
(THREE-DIMENSIONAL MODEL)

Case	Static		Proposed		Faster at
	$t_{10^{-2}}$	$t_{10^{-3}}$	$t_{10^{-2}}$	$t_{10^{-3}}$	10^{-3}
<i>Compact-model order</i> ($V_{\max} = 8\text{ V}$)					
$K = 8$ (24.42 mrad)	111	740	110	5610	0/6
$K = 12$ (3.59 mrad)	111	740	50	740	4/6
$K = 24$ (0.16 mrad)	111	740	58	195	6/6
<i>Driver voltage limit</i> ($K = 24$)					
$V_{\max} = 6.5\text{ V}$	111	740	59	199	6/6
$V_{\max} = 8\text{ V}$	111	740	58	195	6/6
$V_{\max} = 10\text{ V}$	111	740	59	196	6/6
<i>DAC resolution</i> ($K = 24$, $V_{\max} = 8\text{ V}$)					
10 bits (floor 5.8×10^{-3})	111	740	71	1970	2/6
12 bits (floor 1.3×10^{-3})	111	740	59	1440	0/6
14 bits (floor 3.7×10^{-4})	111	740	58	194	6/6
16 bits (floor 8.6×10^{-5})	111	740	58	192	6/6

Settling times in μ s. Model order: largest phase error of the compact model (Table II). DAC: error floor of the quantised static calibration; settling measured to the quantised steady state.

10 V, changed the proposed method's median settling times by less than 2% (Table IV). This agrees with the waveforms of Fig. 4: the settling-time gain comes from the small correction that cancels slow crosstalk, not from overdrive. The method therefore does not require drivers with extra voltage range.

E. Fabrication Variations

A chip never matches its design model exactly. Table V applies the waveforms designed on the nominal model to chips whose stack differs from it: oxide conductivity $\pm 10\%$, buried-oxide thickness 1.8 or 2.2 μ m, heater height $+0.5 \mu$ m and heater resistance $\pm 5\%$. Each chip keeps its own static calibration, as in practice, and receives the nominal dynamic correction on top of it. The proposed drive remained faster than static compensation at both thresholds in all 42 cases (7 chips, 6 reconfigurations each), and its median settling time to 10^{-3} stayed between 186 and 236 μ s, against 662–785 μ s for static compensation. A design-time thermal model thus suffices for these global variations, which change all time constants together. Local, heater-to-heater variations were not studied here; a per-chip identification from measured step responses would address them.

F. Converter Resolution

The corrections are millivolts, so the drive's DAC resolution matters. Table IV quantises all voltages to n bits over 0 V to 8 V. Quantisation already limits static calibration: the quantised static voltages leave a steady matrix error of 1.3×10^{-3} at 12 bits and 3.7×10^{-4} at 14 bits, so a 10^{-3} accuracy requires at least 14 bits whatever the drive. Measured against the quantised steady state, the proposed drive kept its full gain

TABLE V

FABRICATION VARIATIONS: WAVEFORMS DESIGNED ON THE NOMINAL MODEL, APPLIED TO OTHER CHIPS (MEDIAN OF 6 RECONFIGURATIONS)

Chip	Static		Proposed	
	t_{10-2}	t_{10-3}	t_{10-2}	t_{10-3}
Nominal chip	111	740	58	195
Oxide conductivity -10%	107	690	62	210
Oxide conductivity $+10\%$	116	785	58	202
BOX $1.8\mu\text{m}$	109	740	54	195
BOX $2.2\mu\text{m}$	104	662	59	236
Heater height $+0.5\mu\text{m}$	117	770	62	186
Heater resistance -5%	111	740	58	198
Heater resistance $+5\%$	111	740	59	190

Settling times in μs , on each chip's three-dimensional model with its own static calibration.

at 14 and 16 bits. At 12 bits it kept the gain at 10^{-2} but lost it at 10^{-3} , because the slowly decaying correction is then only a few quantisation steps. The method therefore needs no more resolution than the accuracy target itself demands.

VII. DISCUSSION

A. What Limits Reconfiguration

The results separate two time scales. The first tens of microseconds are limited by each heater's own thermal response and by the driver's headroom; pre-emphasis addresses them, and additional coordination adds little there (Fig. 3). Beyond that, the error is dominated by heat exchanged between heaters through the substrate, with time constants up to 2.4 ms. Static compensation fixes where this heat ends up but not when it gets there, and pre-emphasis applied heater by heater cannot see it. Because the slow heat is linear, predictable and small, a correction of a few to a few tens of millivolts per heater, shaped over about a millisecond, cancels most of it. This is why the gain grows with the accuracy required: a factor of about 2 at 10^{-2} and about 4 at 10^{-3} .

B. Practical Implementation

The method changes only the voltage waveforms. Programmable meshes are already driven by multichannel digital-to-analogue converters (DACs), so the waveforms can be stored and replayed for each reconfiguration, or computed on demand in about 100 s on one CPU core. The static calibration that every programmable mesh already needs is kept unchanged as the end point. The dynamic model needs heater step responses at the interval phases. In our workflow they came from a thermal simulation of the layout; on hardware they can be measured once per chip design, or they can be taken from simulation and refined with a few measurements, and Section VI-E indicates that a design-time model is adequate under the global stack variations studied. The corrections are small, so the drive electronics must resolve millivolts: Section VI-F shows the resolution needed. Designs for many reconfigurations can be computed in parallel, and warm-starting from a neighbouring design, which we did not exploit, would shorten design time further.

C. Limitations

This study has limitations that bound its conclusions. (i) All results are simulations. The three-dimensional model is validated against a direct solver, for grid convergence and against published heater data, but not against a measured mesh; the mesh transfer matrix uses ideal couplers and neglects loss and fabrication errors, which static calibration absorbs and which do not change the thermal dynamics. (ii) The drive is open loop. It relies on the dynamic model; Sections VI-C and VI-E quantify how model errors affect it, but drift of the ambient or package temperature over long times is not modelled. (iii) One mesh size and one layout were studied. We expect denser layouts, whose crosstalk is stronger, to benefit more, and layouts with thermal isolation trenches less, but this remains to be quantified. (iv) The design time of about 100 s suits pre-computed reconfigurations, not on-the-fly updates within microseconds.

VIII. CONCLUSION

Thermal crosstalk in thermo-optic programmable photonic meshes is not only a steady-state error but also a slow transient: small, far-reaching and governed by substrate time constants up to milliseconds. A three-dimensional thermal model of an 8×8 silicon-on-insulator Clements mesh, solved exactly by modal separation, shows that this transient, not the heater's own response, sets the time needed to reconfigure the mesh to high accuracy. Static crosstalk compensation and per-heater pre-emphasis leave it essentially untouched. Designing the drive waveforms of all heaters jointly, by gradient-based optimisation of the exact matrix error through a compact shared-pole thermal model, cut the median settling time by a factor of 2.2 to a 10^{-2} matrix error and by a factor of 4.3 to 10^{-3} , in every one of 16 random reconfigurations evaluated on the three-dimensional model. The approach needs no hardware change and keeps the existing static calibration. The code and data are openly available.

CODE AND DATA AVAILABILITY

The thermal model, identification, drive design and evaluation code, and the data behind every result, are openly available at <https://github.com/Arfin005/thermo-optic-crosstalk-drive>.

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